

Note 10: The shor's algorithm

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10.1 Summary of Progress

1. Fully derive all steps of Shor's factoring algorithm.
2. Explain and prove QFT and inverse QFT circuits.
3. Study phase estimation for general and modular unitaries.
4. Construct and analyze the quantum order-finding method.
5. Prove basic mathematical rules and the algorithms success probability.
6. Prove the computational cost and speed advantage of Shor's algorithm.

10.2 The entire algorithm

1. Quantum modules
 - Inverse QFT
 - General unitary phase estimation
 - Modular phase estimation U_a
 - Quantum order-finding
2. Classical modules
 - Continued fraction algorithm
 - Euclidean gcd, modular exponentiation
 - Top-level Shor factorization loop

QFT $\subsetneq$ Modular Phase Estimation $\subsetneq$ Order Finding $\subsetneq$ Shor's Factoring Algorithm

1. Inverse QFT Pseudocode

Algorithm InvQFT(n)**Input:** n qubits register $|x_{n-1}x_{n-2}\dots x_0\rangle$

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1: for  $m = 0$  to  $\lfloor n/2 \rfloor - 1$  do
2:   SWAP( $m, n - 1 - m$ )
3: end for
4: for  $m = 0$  to  $n - 1$  do
5:   for  $l = n - 1 - m$  downto  $1$  do
6:      $\text{CR}_{l+1}^\dagger(\text{ctrl} = m + l, \text{target} = m)$ 
7:   end for
8:    $H(m)$ 
9: end for

```

Gate definition: $R_k^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-2\pi i/2^k} \end{pmatrix}$

2. General Phase Estimation Pseudocode

Algorithm PhaseEst($n, U, |u\rangle$)

Input: Register length n , unitary U , eigenstate $U|u\rangle = e^{2\pi i\varphi}|u\rangle$

Output: Measured integer x

```

1:  $|\Psi\rangle \leftarrow |0\rangle^{\otimes n} \otimes |u\rangle$ 
2: Apply  $H^{\otimes n}$  to phase register
3: for  $j = 0$  to  $n - 1$  do
4:   C- $U^{2^j}$  (ctrl =  $j$ , target = eigen-reg)
5: end for
6: InvQFT( $n$ ) on phase register
7: Measure phase register, get integer  $x$ 
8: return  $x$ 

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3. Continued Fraction Pseudocode

Algorithm ContinuedFrac(α)

Input: $\alpha \in [0, 1)$

Output: All convergents $\{p_k/q_k\}$

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1: Initialize  $p_{-1} = 1, p_0 = 0; q_{-1} = 0, q_0 = 1$ 
2:  $\alpha_k \leftarrow \alpha, \text{Conv} \leftarrow \emptyset$ 
3: while  $\alpha_k \neq 0$  do
4:    $a_k \leftarrow \lfloor \alpha_k \rfloor$ 
5:    $p_k \leftarrow a_k p_{k-1} + p_{k-2}$ 
6:    $q_k \leftarrow a_k q_{k-1} + q_{k-2}$ 
7:   Append  $\frac{p_k}{q_k}$  to Conv
8:    $\alpha_k \leftarrow 1/(\alpha_k - a_k)$ 
9: end while
10: return Conv

```

4. Full Shor Factoring Pseudocode

Algorithm ShorFactor(N)

Input: Odd composite N , not prime power

Output: Nontrivial divisor $d \in (1, N)$

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1: while true do
2:   Sample random  $a \in [2, N - 1]$ 
3:    $r \leftarrow \text{OrderFind}(a, N)$ 
4:   if  $r = \text{Fail}$  or  $r$  odd then
5:     continue
6:   end if
7:    $M \leftarrow a^{r/2} \bmod N, d_+ \leftarrow \gcd(M - 1, N), d_- \leftarrow \gcd(M + 1, N)$ 
8:   if  $1 < d_+ < N$  then
9:     return  $d_+$ 
10:  end if
11:  if  $1 < d_- < N$  then
12:    return  $d_-$ 
13:  end if
14: end while

```

5. Quantum Order-Finding Pseudocode

Algorithm OrderFind(a, N)

Input: Coprime integers a, N

Output: Order r or Fail

```

1: while true do
2:    $(s_1, r_1) \leftarrow \text{SingleRun}(a, N)$ 
3:    $(s_2, r_2) \leftarrow \text{SingleRun}(a, N)$ 
4:   if  $(s_1, r_1) = \text{Fail}$  or  $(s_2, r_2) = \text{Fail}$  then
5:     continue
6:   end if
7:    $r_* \leftarrow \text{lcm}(r_1, r_2)$ 
8:   if  $a^{r_*} \equiv 1 \pmod{N}$  then
9:     return  $r_*$ 
10:  end if
11: end while

```

Subroutine SingleRun(a, N)

```

1: Set register width  $n$ 
2: Construct modular unitary  $U_a : |y\rangle \mapsto |ay \bmod N\rangle$ 
3:  $x \leftarrow \text{PhaseEst}(n, U_a, |1\rangle)$ 
4:  $\alpha \leftarrow x/2^n$ 
5:  $\text{ConvList} \leftarrow \text{ContinuedFrac}(\alpha)$ 
6: for  $\frac{s_1}{r_1} \in \text{ConvList}$  do
7:   return  $(s_1, r_1)$ 
8: end for
9: return Fail

```

10.3 Circuit of QFT

Definition 1 The quantum Fourier transform (QFT) on an orthonormal basis $\{|0\rangle, \dots, |N-1\rangle\}$ is defined to be a linear operator $\mathcal{F}$ acting as

$$\mathcal{F}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi\sqrt{-1}jk/N} |k\rangle.$$

Remark 2 $\mathcal{F}$ is a unitary operator, thus $\mathcal{F}^{-1} = \mathcal{F}^\dagger$. Replace $\sqrt{-1}$ with $-\sqrt{-1}$ in the formula, and we obtain the inverse Fourier transform $\mathcal{F}^\dagger$.

$$\mathcal{F}^\dagger|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{-2\pi\sqrt{-1}jk/N} |k\rangle$$

Lemma 3 Let $N = 2^n$, $j \in \{0, 1, \dots, 2^n - 1\}$ with binary expansion $j = j_{n-1}j_{n-2} \dots j_0$. Then

$$\mathcal{F}|j\rangle = \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left(|0\rangle + e^{\sqrt{-1}2\pi j/2^l} |1\rangle \right) = \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left(|0\rangle + e^{\sqrt{-1}2\pi \cdot 0.j_{l-1}j_{l-2} \dots j_0} |1\rangle \right).$$

Proof

$$\begin{aligned}
\mathcal{F}|j\rangle &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{\sqrt{-1} 2\pi j k / 2^n} |k\rangle = \frac{1}{\sqrt{2^n}} \sum_{k_0, \dots, k_{n-1}=0}^1 e^{\sqrt{-1} 2\pi j \sum_{l=0}^{n-1} k_l / 2^{n-l}} \bigotimes_{l=0}^{n-1} |k_l\rangle \\
&= \frac{1}{\sqrt{2^n}} \bigotimes_{l=0}^{n-1} \sum_{k_l=0}^1 e^{\sqrt{-1} 2\pi j k_l / 2^{n-l}} |k_l\rangle = \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left(|0\rangle + e^{\sqrt{-1} 2\pi j / 2^l} |1\rangle \right) \\
&= \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left(|0\rangle + e^{\sqrt{-1} 2\pi \cdot 0 \cdot j_{l-1} j_{l-2} \dots j_0} |1\rangle \right) \\
&= \frac{1}{\sqrt{2^n}} \left(|0\rangle + e^{\sqrt{-1} 2\pi \cdot 0 \cdot j_0} |1\rangle \right) \otimes \left(|0\rangle + e^{\sqrt{-1} 2\pi \cdot 0 \cdot j_1 j_0} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{\sqrt{-1} 2\pi \cdot 0 \cdot j_{n-1} j_{n-2} \dots j_0} |1\rangle \right).
\end{aligned}$$

■

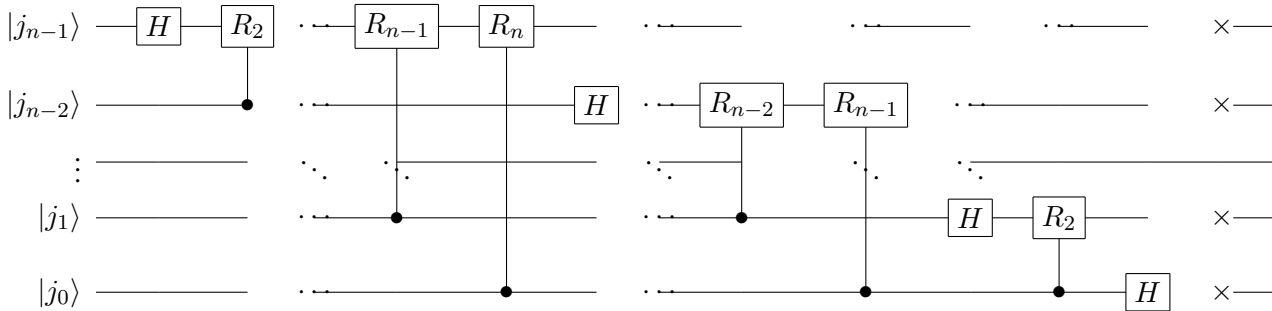
Theorem 4 The quantum Fourier transform $\mathcal{F}$ on n qubits ($N = 2^n$) can be decomposed as a product of Hadamard gates and controlled-phase gates.

- Let the phase gate $R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi\sqrt{-1}/2^k} \end{pmatrix}$.
- $CR_k^{(a,b)}$ is the controlled- R_k gate with control qubit a and target qubit b ,
- $\text{SWAP}_{a,b}$ swaps qubits a, b .

Then the overall operator factorization of the n -qubit QFT reads

$$\mathcal{F} = \left(\prod_{t=0}^{\lfloor n/2 \rfloor - 1} \text{SWAP}_{t, n-1-t} \right) \prod_{m=0}^{n-1} \left(H_m \prod_{l=1}^{n-1-m} CR_{l+1}^{(m, m+l)} \right),$$

whose corresponding quantum circuit is drawn below.



Proof By the preceding lemma, the QFT output factorizes as

$$\mathcal{F}|j\rangle = \frac{1}{\sqrt{2^n}} \bigotimes_{l=1}^n \left(|0\rangle + e^{\sqrt{-1} 2\pi \cdot 0 \cdot j_{l-1} \dots j_0} |1\rangle \right).$$

Take arbitrary qubit m , input state $|j_m\rangle$. First apply Hadamard gate H_m :

$$H|j_m\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\sqrt{-1} \cdot 2\pi \cdot \frac{j_m}{2}} |1\rangle \right).$$

Introduce controlled- R_{l+1} with control qubit $m+l$ ($l \geq 1$), target qubit m . The definition of R_k gives

$$CR_{l+1}^{(m, m+l)} |j_{m+l}\rangle \otimes |\psi_m\rangle = |j_{m+l}\rangle \otimes R_{l+1}^{j_{m+l}} |\psi_m\rangle, \quad R_{l+1} = \begin{pmatrix} 1 & 0 \\ 0 & e^{\sqrt{-1} \cdot 2\pi / 2^{l+1}} \end{pmatrix}.$$

Act $CR_{l+1}^{(m,m+l)}$ on $H|j_m\rangle$:

$$CR_{l+1}^{(m,m+l)} H|j_m\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\sqrt{-1} \cdot 2\pi \left(\frac{j_m}{2} + \frac{j_{m+l}}{2^{l+1}} \right)} |1\rangle \right).$$

Iterate over all $l = 1, 2, \dots, n-1-m$:

$$\prod_{l=1}^{n-1-m} CR_{l+1}^{(m,m+l)} H_m |j_m\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{\sqrt{-1} \cdot 2\pi \cdot 0 \cdot j_m j_{m+1} \dots j_{n-1}} |1\rangle \right).$$

Process every qubit $m = n-1, n-2, \dots, 0$ in sequence, the overall state before swap gates is

$$\bigotimes_{m=0}^{n-1} \left(\prod_{l=1}^{n-1-m} CR_{l+1}^{(m,m+l)} H_m \right) |j_m\rangle = \frac{1}{\sqrt{2^n}} \bigotimes_{m=0}^{n-1} \left(|0\rangle + e^{\sqrt{-1} \cdot 2\pi \cdot 0 \cdot j_m j_{m+1} \dots j_{n-1}} |1\rangle \right).$$

This tensor order is reversed compared with the lemmas expression. Applying swap gates $\text{SWAP}_{t,n-1-t}$ for $t = 0, 1, \dots, \lfloor n/2 \rfloor - 1$ flips the qubit sequence, matching the required tensor product form. Concatenate all unitary operators to obtain the factorization

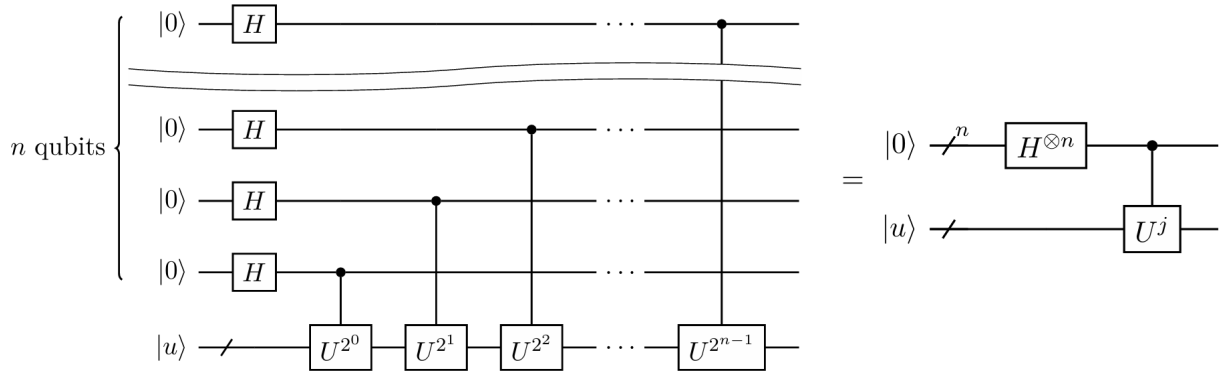
$$\mathcal{F} = \left(\prod_{t=0}^{\lfloor n/2 \rfloor - 1} \text{SWAP}_{t,n-1-t} \right) \prod_{m=0}^{n-1} \left(H_m \prod_{l=1}^{n-1-m} CR_{l+1}^{(m,m+l)} \right).$$

■

Remark 5 Replace $\sqrt{-1}$ with $-\sqrt{-1}$ in the formula, and we obtain the inverse Fourier transform $\mathcal{F}^\dagger$. R_k is replaced by

$$R_k^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-2\pi\sqrt{-1}/2^k} \end{pmatrix}.$$

10.4 Phase estimation for Unitary operator



Theorem 6 Let U be a unitary operator and vector $|u\rangle \neq 0$ satisfying

$$U|u\rangle = e^{2\pi\sqrt{-1}\varphi}|u\rangle, \quad \varphi \in [0, 1).$$

1. Circuit consisting of an n -qubit register initialized to $|0\rangle^{\otimes n}$ and an auxiliary qubit $|u\rangle$;
2. Apply $H^{\otimes n}$ to the n -qubit register;
3. Apply controlled- U^{2^j} gates for $j = 0, 1, \dots, n-1$, where the j -th gate is controlled by the j -th qubit of the register.

After the above operations, the joint state of the two registers is

$$\frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}\varphi k} |k\rangle \otimes |u\rangle.$$

Proof Start with the initial separable state $|0\rangle^{\otimes n} \otimes |u\rangle$. First apply the tensor product of Hadamard gates $H^{\otimes n} \otimes \mathbb{I}$ to the n -qubit register:

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle,$$

so the intermediate state reads

$$(H^{\otimes n} \otimes \mathbb{I})(|0\rangle^{\otimes n} \otimes |u\rangle) = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle \otimes |u\rangle.$$

Write the binary expansion $k = \sum_{j=0}^{n-1} k_{n-1-j} 2^{n-1-j}$ with $k_t \in \{0, 1\}$. The controlled- U^{2^j} operation acts as $C(U^{2^j})|k\rangle|u\rangle = |k\rangle(U^{2^j})^{k_j}|u\rangle$. Applying all $C(U^{2^0}), C(U^{2^1}), \dots, C(U^{2^{n-1}})$ sequentially:

$$\begin{aligned} \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle|u\rangle &\xrightarrow{\prod_{j=0}^{n-1} C(U^{2^j})} \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle [U^{2^{n-1}}]^{k_{n-1}} \dots [U^{2^1}]^{k_1} [U^{2^0}]^{k_0} |u\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle e^{2\pi\sqrt{-1}\varphi 2^{n-1}k_{n-1}} \dots e^{2\pi\sqrt{-1}\varphi 2^1k_1} e^{2\pi\sqrt{-1}\varphi 2^0k_0} |u\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle e^{2\pi\sqrt{-1}\varphi \sum_{j=0}^{n-1} 2^j k_j} |u\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}\varphi k} |k\rangle \otimes |u\rangle. \end{aligned}$$

This completes the proof. ■

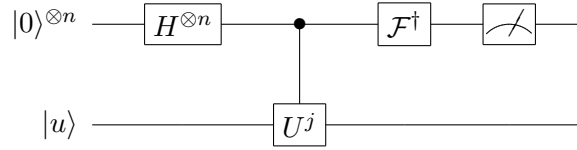
Lemma 7 Let $U|u\rangle = e^{2\pi\sqrt{-1}\varphi}|u\rangle$ with $\varphi \in [0, 1)$. After the phase estimation controlled- U^{2^j} gates, apply inverse quantum Fourier transform $\mathcal{F}^\dagger$ on the n -bit register. The state of the n -qubit register is

$$\mathcal{F}^\dagger \left(\frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}\varphi k} |k\rangle \right) = \frac{1}{2^n} \sum_{j=0}^{2^n-1} \sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}k(\varphi - \frac{j}{2^n})} |j\rangle.$$

Proof By definition of inverse QFT $\mathcal{F}^\dagger|k\rangle = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{-2\pi\sqrt{-1}kj/2^n} |j\rangle$.

$$\begin{aligned} \mathcal{F}^\dagger \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}\varphi k} |k\rangle &= \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}\varphi k} \cdot \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{-2\pi\sqrt{-1}kj/2^n} |j\rangle \\ &= \frac{1}{2^n} \sum_{j=0}^{2^n-1} \left(\sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}k(\varphi - \frac{j}{2^n})} \right) |j\rangle. \end{aligned}$$
■

Consider the circuit



Theorem 8 Let $U|u\rangle = e^{2\pi\sqrt{-1}\varphi}|u\rangle$ with $\varphi \in [0, 1)$. If

- The input state of phase estimation is $|0\rangle^{\otimes n} \otimes |u\rangle$.
- The phase register (register 1) is measured under $\{|x\rangle : x \in \{0, 1\}^n\}$,
- The outcome corresponds to integer $2^n \tilde{\varphi}$.

The probability of measuring outcome $|2^n \tilde{\varphi}\rangle$ in the n -qubit phase register satisfies

$$\mathbb{P}(2^n \tilde{\varphi}) = \left| \frac{\sin(2^n \pi(\varphi - \tilde{\varphi}))}{2^n \sin(\pi(\varphi - \tilde{\varphi}))} \right|^2.$$

Let $\tilde{\varphi}_{\text{opt}}$ be the optimal n -bit approximation of φ with $|\varphi - \tilde{\varphi}_{\text{opt}}| \leq \frac{1}{2^{n+1}}$, then

$$\mathbb{P}(|\varphi - \tilde{\varphi}| \leq \frac{1}{2^{n+1}}) = \mathbb{P}(2^n \tilde{\varphi}_{\text{opt}}) \geq \frac{4}{\pi^2} \approx 0.405.$$

Proof From Theorem 1, the probability of outcome $|2^n \tilde{\varphi}\rangle$ is the squared modulus of the coefficient of $|j = 2^n \tilde{\varphi}\rangle$:

$$\mathbb{P}(2^n \tilde{\varphi}) = \left| \frac{1}{2^n} \sum_{k=0}^{2^n-1} \left[e^{2\pi\sqrt{-1}(\varphi - \tilde{\varphi})} \right]^k \right|^2.$$

Evaluate the finite geometric series $\sum_{k=0}^{M-1} x^k = \frac{1-x^M}{1-x}$ with $M = 2^n$, $x = e^{2\pi\sqrt{-1}(\varphi - \tilde{\varphi})}$:

$$\sum_{k=0}^{2^n-1} e^{2\pi\sqrt{-1}k(\varphi - \tilde{\varphi})} = \frac{1 - e^{2\pi\sqrt{-1}2^n(\varphi - \tilde{\varphi})}}{1 - e^{2\pi\sqrt{-1}(\varphi - \tilde{\varphi})}}.$$

Factor out phase terms and use identity $|1 - e^{\sqrt{-1}\theta}| = 2|\sin(\theta/2)|$:

$$\begin{aligned} \left| \frac{1 - e^{2\pi\sqrt{-1}2^n(\varphi - \tilde{\varphi})}}{1 - e^{2\pi\sqrt{-1}(\varphi - \tilde{\varphi})}} \right| &= \left| \frac{e^{\sqrt{-1}\pi 2^n(\varphi - \tilde{\varphi})} \cdot 2 \sin(2^n \pi(\varphi - \tilde{\varphi}))}{e^{\sqrt{-1}\pi(\varphi - \tilde{\varphi})} \cdot 2 \sin(\pi(\varphi - \tilde{\varphi}))} \right| \\ &= \frac{|\sin(2^n \pi(\varphi - \tilde{\varphi}))|}{|\sin(\pi(\varphi - \tilde{\varphi}))|}. \end{aligned}$$

Substitute back to recover the probability formula:

$$\mathbb{P}(2^n \tilde{\varphi}) = \left| \frac{\sin(2^n \pi(\varphi - \tilde{\varphi}))}{2^n \sin(\pi(\varphi - \tilde{\varphi}))} \right|^2.$$

For the optimal n -bit approximation $\tilde{\varphi}_{\text{opt}}$, we have $|\varphi - \tilde{\varphi}_{\text{opt}}| \leq \frac{1}{2^{n+1}}$. Use the bound $|\sin(x)| \geq \frac{2|x|}{\pi}$ for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$:

$$|\sin(2^n \pi(\varphi - \tilde{\varphi}_{\text{opt}}))| \geq \frac{2}{\pi} \cdot 2^n \pi |\varphi - \tilde{\varphi}_{\text{opt}}| = 2^{n+1} |\varphi - \tilde{\varphi}_{\text{opt}}|.$$

Also $|\sin(\pi(\varphi - \tilde{\varphi}_{\text{opt}}))| \leq \pi |\varphi - \tilde{\varphi}_{\text{opt}}|$. Plug into the probability expression:

$$\mathbb{P}(2^n \tilde{\varphi}_{\text{opt}}) \geq \left| \frac{2^{n+1}(\varphi - \tilde{\varphi}_{\text{opt}})}{2^n \pi(\varphi - \tilde{\varphi}_{\text{opt}})} \right|^2 = \left(\frac{2}{\pi} \right)^2 = \frac{4}{\pi^2} \approx 0.405.$$

■

Corollary 9 Let $\{U|u\rangle = e^{2\pi i\varphi_u}|u\rangle\}$ be all eigenpairs of U , $\varphi_u \in [0, 1)$.

Replace the input $|0\rangle^{\otimes n} \otimes |u\rangle$ **with** $|0\rangle^{\otimes n} \otimes |\psi\rangle$ **for any normalized state** $|\psi\rangle$. We have

$$\mathbb{P}(\exists u, |\varphi_u - \tilde{\varphi}| \leq \frac{1}{2^{n+1}}) \geq \frac{4}{\pi^2} \approx 0.405.$$

Proof Let $|\psi\rangle = \sum_u c_u |u\rangle$. By linearity of phase estimation, the output state is

$$\sum_u c_u \cdot |\tilde{\varphi}_u\rangle \otimes |u\rangle,$$

From the preceding theorem, for each fixed u ,

$$\mathbb{P}\left(|\varphi_u - \tilde{\varphi}_u| \leq \frac{1}{2^{n+1}} \middle| u\right) \geq \frac{4}{\pi^2}.$$

The total probability that the measured $\tilde{\varphi}$ is a qualified approximation for some φ_u satisfies

$$\mathbb{P}(\exists u, |\varphi_u - \tilde{\varphi}| \leq \frac{1}{2^{n+1}}) = \sum_u \mathbb{P}(u) \mathbb{P}(|\varphi_u - \tilde{\varphi}_u| \leq \frac{1}{2^{n+1}} | u) \geq \sum_u |c_u|^2 \cdot \frac{4}{\pi^2} = \frac{4}{\pi^2}.$$

■

10.5 Phase estimation for $U_a : |x\rangle \mapsto |ax \bmod N\rangle$

Lemma 10 If $\gcd(a, N) = 1$, then the modular multiplication operator defined by

$$U_a|y\rangle = \begin{cases} |ay \bmod N\rangle & 0 \leq y < N \\ |y\rangle & y \geq N \end{cases}$$

is a unitary operator.

Proof We have $\{U_a : \gcd(a, N) = 1\} \simeq (\mathbb{Z}/N\mathbb{Z})^\times$ as group, so U_a is a permutation of ONS-basis $|0\rangle, |1\rangle, \dots, |2^L - 1\rangle$. Thus U_a is unitary.

In addition, we have $U^r = I$, so there is $U_a|u_s\rangle = e^{2\pi i s/r}|u_s\rangle$ for some $s \in \mathbb{Z}$ and $|u_s\rangle \neq 0$.

■

Primitive gate assumption for phase estimation circuit

Abstract phase estimation relies on the existence of controlled unitary gates $C-U^{2^j}$ for each $j = 0, 1, \dots, n-1$. For our modular multiplication case $U = U_a$, we adopt the identical formal assumption first:

Assumption: For every integer $j \geq 0$, the controlled modular multiplication gate $C - U_a^{2^j}$ is accessible as an elementary quantum gate. Its action is

$$C - U_a^{2^j} |c\rangle |y\rangle = \begin{cases} |c\rangle |a^{2^j} y \bmod N\rangle & c = 1, \\ |c\rangle |y\rangle & c = 0. \end{cases}$$

This is a temporary black-box assumption inherited from general phase estimation. Unlike abstract U , $U_a^{2^j}$ has a concrete number-theoretic form $a^{2^j} \pmod{N}$, so we may further implement this gate classically via repeated modular squaring, then map the arithmetic circuit to quantum reversible gates.

Remain Problem: This work takes $C - U_a^{2^j}$ as a primitive gate without explicit decomposition. Future research shall construct full universal-gate circuits for U_a and its controlled powers, together with their physical Hamiltonian realization schemes.

10.6 Order Finding

Section-wide Global Assumption Let x denote the integer measurement outcome of the n -qubit phase register, define the n -bit **optimal binary phase approximation**

$$\tilde{\varphi} = \frac{x}{2^n},$$

which satisfies the uniform error bound $|\varphi - \tilde{\varphi}| \leq \frac{1}{2^{n+1}}$ for the **true phase** φ . All subsequent analysis and results rely on this definition and error estimate of $\tilde{\varphi}$.

Lemma 11 Let $\alpha \in \mathbb{R}$, and suppose rational numbers $\frac{a}{b}, \frac{c}{d}$ satisfy

$$\left| \alpha - \frac{a}{b} \right| < \frac{1}{2b^2}, \quad \left| \alpha - \frac{c}{d} \right| < \frac{1}{2d^2}, \quad b, d > 0$$

Then $\frac{a}{b} = \frac{c}{d}$.

Proof By contradiction. Assume $\frac{a}{b} \neq \frac{c}{d}$. Since the fractions are in lowest terms,

$$\left| \frac{a}{b} - \frac{c}{d} \right| = \frac{|ad - bc|}{bd} \geq \frac{1}{bd}.$$

By triangle inequality:

$$\left| \frac{a}{b} - \frac{c}{d} \right| \leq \left| \alpha - \frac{a}{b} \right| + \left| \alpha - \frac{c}{d} \right| < \frac{1}{2b^2} + \frac{1}{2d^2}.$$

Combine two inequalities:

$$\frac{1}{bd} < \frac{b^2 + d^2}{2b^2d^2} \implies 2bd < b^2 + d^2 \implies (b - d)^2 < 0.$$

This contradiction implies $\frac{a}{b} = \frac{c}{d}$. ■

Corollary 12 Let r be the order of a with $r < 2^{n/2}$. Any positive integer s', r' satisfying

$$\gcd(s', r') = 1, \quad \left| \tilde{\varphi} - \frac{s'}{r'} \right| < \frac{1}{2^{n+1}}$$

Then $s'/r' = s/r$. If $\gcd(s, r) = 1$, then r' coincides with the order of a , i.e., $r' = r$.

Proof We have $\varphi = \frac{s}{r}$ as the true phase, with global error bound

$$|\varphi - \tilde{\varphi}| \leq \frac{1}{2^{n+1}}.$$

By triangle inequality:

$$\left| \frac{s}{r} - \frac{s'}{r'} \right| \leq \left| \frac{s}{r} - \tilde{\varphi} \right| + \left| \tilde{\varphi} - \frac{s'}{r'} \right| < \frac{1}{2^{n+1}} + \frac{1}{2^{n+1}} = \frac{1}{2^n}.$$

Since $r < 2^{n/2}$ and we only consider candidate $r' < 2^{n/2}$, we have $rr' < 2^n$. If $\frac{s}{r} \neq \frac{s'}{r'}$,

$$\left| \frac{s}{r} - \frac{s'}{r'} \right| = \frac{|sr' - s'r|}{rr'} \geq \frac{1}{rr'} > \frac{1}{2^n},$$

which contradicts the previous bound. Thus $\frac{s}{r} = \frac{s'}{r'}$ and $r' = r$. ■

Mathematical Theory of Continued Fraction

Definition 13 (continued fraction) Given a real number α , its simple continued fraction expansion is written as

$$\alpha = [a_0; a_1, a_2, \dots, a_k] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_k}}}}.$$

Theorem 14 Denote the k -th convergent as $\frac{p_k}{q_k}$. The convergents satisfy the recurrence formulas:

$$\begin{cases} p_{-1} = 1, & p_0 = a_0, & p_k = a_k p_{k-1} + p_{k-2}, \\ q_{-1} = 0, & q_0 = 1, & q_k = a_k q_{k-1} + q_{k-2}. \end{cases}$$

Key Identity

$$p_{k-1}q_k - p_kq_{k-1} = (-1)^k.$$

Proof We proceed by mathematical induction.

Firstly, we prove the recurrence formulas. If it's true for $k = n$, then for $k = n + 1$, we have

$$\begin{aligned} \alpha_{n+1} &= [a_0; a_1, \dots, a_n, a_{n+1}] = \left[a_0; a_1, \dots, a_n + \frac{1}{a_{n+1}} \right] \\ &= \frac{\tilde{p}_n}{\tilde{q}_n} = \frac{\left(a_n + \frac{1}{a_{n+1}} \right) p_{n-1} + p_{n-2}}{\left(a_n + \frac{1}{a_{n+1}} \right) q_{n-1} + q_{n-2}} \\ &= \frac{\left(a_n + \frac{1}{a_{n+1}} \right) p_{n-1} + p_{n-2}}{\left(a_n + \frac{1}{a_{n+1}} \right) q_{n-1} + q_{n-2}} = \frac{(a_n a_{n+1} + 1) p_{n-1} + a_{n+1} p_{n-2}}{(a_n a_{n+1} + 1) q_{n-1} + a_{n+1} q_{n-2}} \\ &= \frac{a_{n+1} (a_n p_{n-1} + p_{n-2}) + p_{n-1}}{a_{n+1} (a_n q_{n-1} + q_{n-2}) + q_{n-1}} = \frac{a_{n+1} p_n + p_{n-1}}{a_{n+1} q_n + q_{n-1}} \\ &= \frac{p_{n+1}}{q_{n+1}} \end{aligned}$$

So it is true for $k = n + 1$.

Now, let's prove the identity $p_{k-1}q_k - p_kq_{k-1} = (-1)^k$.

- Base case $k = 0$:

$$p_{-1}q_0 - p_0q_{-1} = 1 \cdot 1 - a_0 \cdot 0 = 1 = (-1)^0.$$

- Inductive hypothesis: Assume $p_{m-1}q_m - p_mq_{m-1} = (-1)^m$ holds for $k = m$.

- Inductive step for $k = m + 1$:

$$\begin{aligned} p_m q_{m+1} - p_{m+1} q_m &= p_m (a_{m+1} q_m + q_{m-1}) - (a_{m+1} p_m + p_{m-1}) q_m \\ &= p_m q_{m-1} - p_{m-1} q_m \\ &= -(-1)^m = (-1)^{m+1}. \end{aligned}$$

The identity holds for all non-negative integers k . ■

Corollary 15 The fraction $\frac{p_k}{q_k}$ is reduced. In other words, $\gcd(p_k, q_k) = 1$.

Theorem 16 Let α be a rational number and suppose p/q is a rational number such that

$$\left| \frac{p}{q} - \alpha \right| \leq \frac{1}{2q^2}.$$

Then p/q is a convergent of the continued fraction for α .

Proof Let $p/q = [a_0, \dots, a_n]$ be the continued fraction expansion for p/q , and define p_j, q_j as in Theorem 14, so that $p_n/q_n = p/q$. Since $\gcd(p_n, q_n) = 1$, $q_n \leq q$. Define δ by the equation

$$\alpha \equiv \frac{p_n}{q_n} + \frac{\delta}{2q_n^2}, \quad \delta = 2q_n^2\alpha - 2p_nq_n$$

so $|\delta| \leq \frac{2q_n^2}{2q^2} \leq 1$. Define λ by the equation

$$\begin{aligned} \lambda &\equiv 2 \left(\frac{q_n p_{n-1} - p_n q_{n-1}}{\delta} \right) - \frac{q_{n-1}}{q_n} \\ &= 2 \left(\frac{q_n p_{n-1} - p_n q_{n-1}}{2q_n^2\alpha - 2p_nq_n} \right) - \frac{q_{n-1}}{q_n} = \frac{q_n p_{n-1} - p_n q_{n-1}}{q_n^2\alpha - p_nq_n} - \frac{q_{n-1}}{q_n} \\ &= \frac{q_n p_{n-1} - p_n q_{n-1} - (q_n\alpha - p_n) q_{n-1}}{q_n (q_n\alpha - p_n)} = \frac{q_n p_{n-1} - q_n q_{n-1}\alpha}{q_n (q_n\alpha - p_n)} \\ &= \frac{p_{n-1} - q_{n-1}\alpha}{q_n\alpha - p_n} \end{aligned}$$

The reason we define λ this way is because with a little algebra we can see that it satisfies the equation

$$\alpha = \frac{\lambda p_n + p_{n-1}}{\lambda q_n + q_{n-1}},$$

and therefore $x = [a_0, \dots, a_n, \lambda]$.

Note that $q_n p_{n-1} - p_n q_{n-1} = (-1)^n$, choosing n even, then

$$\lambda = \frac{2}{\delta} - \frac{q_{n-1}}{q_n}.$$

By the increasing property of q_n it follows that

$$\lambda = \frac{2}{\delta} - \frac{q_{n-1}}{q_n} \geq 2 - 1 \geq 1.$$

Therefore λ is a rational number greater than 1, and so has a simple finite continued fraction, $\lambda = [b_0, \dots, b_m]$, and so $x = [a_0, \dots, a_n, b_0, \dots, b_m]$ is a simple finite continued fraction for x with p/q as a convergent. ■

Remark 17 This theorem implies every convergent is an optimal Diophantine approximation of α . When recovering the order r from $\tilde{\varphi} = x/2^n$, we only need to enumerate convergents of $\tilde{\varphi}$ as candidate fractions, since no other rational numbers can satisfy the required approximation accuracy.

Handling the Case $\gcd(s, r) \neq 1$.

Suppose we execute phase estimation and continued fraction reconstruction twice independently, yielding two pairs (s_1, r_1) and (s_2, r_2) such that $\gcd(s_1, r_1) = \gcd(s_2, r_2) = 1$. Then we take $r_* = \text{lcm}(r_1, r_2)$, and verify whether $a^{r_*} = 1 \pmod{N}$. If the congruence fails, repeat the whole process again.

Lemma 18 *If $a, b, p, q \in \mathbb{N}^*$ such that $\frac{a}{b} = \frac{a_1}{b_1}$ and $\gcd(p, q) = 1$, then $a_1 | a$, $b_1 | b$.*

Proof For $\frac{a}{b} = \frac{p}{q}$, we have $\frac{a}{a_1} = \frac{b}{b_1} = \frac{p}{q}$, and $\gcd(p, q) = 1$. If $q > 1$, then

$$pa_1 = qa, pb_1 = qb \Rightarrow \gcd(pa_1, pb_1) = \gcd(qa, qb)$$

$$p = \text{pgcd}(a_1, b_1) = \gcd(qa, qb) = q\gcd(a, b)$$

This contradicts to $\gcd(p, q) = 1$. So $q = 1 \Rightarrow a/a_1 = b/b_1 = p \in \mathbb{Z}$. ■

Theorem 19 *Suppose we execute phase estimation and continued fraction reconstruction twice independently, yielding two pairs (s_1, r_1) and (s_2, r_2) such that $\gcd(s_1, r_1) = \gcd(s_2, r_2) = 1$. Let r denote the true multiplicative order. Then*

$$\mathbb{P}(r = \text{lcm}(r_1, r_2)) \geq \frac{7}{18}.$$

Proof Suppose $\frac{s_1}{r_1} = \frac{s'}{r}$, $\frac{s_2}{r_2} = \frac{s''}{r}$. By the lemma, we have $s' = ks_1, r = kr_1$ and $s'' = ls_2, r = lr_1$ for some $k, l \in \mathbb{Z}$. Thus $r_1 | r, r_2 | r \Rightarrow r = d\text{lcm}(r_1, r_2)$ for some $d \in \mathbb{Z}$.

If $\gcd(s', s'') = 1$, then d must be equal to 1. Otherwise

$$dr_1 | r = kr_1, dr_2 | r = lr_2 \Rightarrow d | k, l \Rightarrow d | \gcd(ks_1, ls_2) = \gcd(s', s'') = 1$$

So $r = \text{lcm}(r_1, r_2)$. According to this fact, we have

$$\mathbb{P}(r = \text{lcm}(r_1, r_2)) \geq \mathbb{P}(\gcd(s', s'') = 1) \geq 1 - \mathbb{P}(\gcd(s', s'') \neq 1)$$

The true numerator s is uniformly distributed over $\{0, 1, \dots, r-1\}$, i.e. $s', s'' \sim U(\{1, \dots, r\})$. For any prime q , the probability that q divides a uniformly random integer satisfies $\mathbb{P}(q | s') \leq 1/q$, and identically $\mathbb{P}(q | s'') \leq 1/q$. Substitute these bounds:

$$\begin{aligned} \mathbb{P}(\gcd(s', s'') \neq 1) &\leq \sum_q \mathbb{P}(q | s') \mathbb{P}(q | s'') \leq \sum_q \frac{1}{q^2} = \frac{1}{4} + \frac{1}{9} + \sum_{q \geq 5} \frac{1}{q^2} \\ &\leq \frac{13}{36} + \sum_{n \geq 5} \frac{1}{n^2} \leq \frac{13}{36} + \sum_{n \geq 5} \left(\frac{1}{n-1} - \frac{1}{n} \right) = \frac{11}{18} \end{aligned}$$

Thus

$$\mathbb{P}(r = \text{lcm}(r_1, r_2)) \geq 1 - \frac{11}{18} = \frac{7}{18}. \quad \text{■}$$

Remark 20 Since the success probability is bounded below by a constant $7/18$, only $O(1)$ of repetitions are necessary.

10.7 Shor's Algorithm for $N = pq$

The idea is base on the lemma.

Lemma 21 *Let $N = pq$ be the product of two distinct odd primes, and $a \in \mathbb{Z}$ satisfy $\gcd(a, N) = 1$. Let r be the multiplicative order of a modulo N , i.e., the minimal positive integer with $a^r \equiv 1 \pmod{N}$. If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then*

$$\gcd(a^{r/2} - 1, N) > 1, \quad \gcd(a^{r/2} + 1, N) > 1,$$

and both values yield non-trivial divisors of N .

Proof From $a^r \equiv 1 \pmod{N}$ and r is even, we have

$$(a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod{pq}.$$

Since p, q are odd primes, $a^{r/2} \equiv 1$ and $a^{r/2} \equiv -1$ cannot hold modulo the same prime.

By condition $a^{r/2} \not\equiv -1 \pmod{N}$, one prime divides $a^{r/2} - 1$ and the other divides $a^{r/2} + 1$. Thus both $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$ produce non-trivial factors of N . ■

Algorithm Steps

Given composite $N = pq$ with distinct odd primes p, q :

1. Randomly sample integer $a \in [2, N - 1]$.
2. Use quantum order-finding subroutine $\text{OrderFind}(a, N)$ to compute the multiplicative order r of a modulo N .
3. If r fails to be found or r is odd, return to Step 1 and resample a .
4. Compute $M = a^{r/2} \pmod{N}$. If $M \equiv -1 \pmod{N}$, return to Step 1.
5. Calculate $d_+ = \gcd(M - 1, N)$ and $d_- = \gcd(M + 1, N)$. Both d_+, d_- are non-trivial factors of N .

The cost of Euclidean algorithm

1. Fibonacci sequence: $F_1 = 1, F_2 = 1, F_n = F_{n-1} + F_{n-2}$ for all $n \geq 3$.
2. Iteration count of Euclidean algorithm: For integers $a > b > 0$, $T(a, b)$ denotes the total number of division steps. A single division step is defined as $a = qb + r$ with $0 \leq r < b$. The algorithm terminates if $r = 0$; otherwise we recursively compute $\gcd(b, r)$.

Theorem 22 (Lamé's Theorem) *Let integers satisfy $a > b > 0$. If computing $\gcd(a, b)$ requires exactly k division steps, then*

$$a \geq F_{k+2}, \quad b \geq F_{k+1}.$$

Proof We proceed by mathematical induction on the number of steps k .

Base case $k = 1$ holds: If $T(a, b) = 1$, then $b \mid a$ and $a = qb$ with quotient $q \geq 2$ (since $a > b$). We need to verify $a \geq F_3, b \geq F_2$. By definition $F_2 = 1, F_3 = 2$.

- Since $b > 0$, we have $b \geq 1 = F_2$;
- $a \geq 2b \geq 2 \cdot 1 = 2 = F_3$.

Inductive hypothesis Assume the statement holds for all inputs with exactly m division steps: For any $x > y > 0$, if $T(x, y) = m$, then $x \geq F_{m+2}$ and $y \geq F_{m+1}$.

Inductive step: prove for $k = m + 1$ Take $a > b > 0$ such that $T(a, b) = m + 1$. Perform the first division step:

$$a = qb + r, \quad 0 < r < b.$$

Note $r \neq 0$, otherwise the algorithm terminates after one step, contradicting $T(a, b) = m + 1$. After removing the first step, the subproblem $\gcd(b, r)$ has iteration count exactly m :

$$T(b, r) = m.$$

Apply the inductive hypothesis to (b, r) :

$$b \geq F_{m+2}, \quad r \geq F_{m+1}.$$

Since $a > b$, the quotient $q \geq 1$.

$$a = bq + r \geq b + r \geq F_{m+2} + F_{m+1} = F_{m+3}.$$

We already have $b \geq F_{m+2}$. For step count $k = m + 1$, the target bounds are

$$a \geq F_{(m+1)+2} = F_{m+3}, \quad b \geq F_{(m+1)+1} = F_{m+2},$$

which match the derived inequalities exactly.

By mathematical induction, Lamé's Theorem holds for all positive integers k . ■

Theorem 23 (Golden Ratio Iteration Bound) Let $\varphi = \frac{1 + \sqrt{5}}{2}$ be the golden ratio. For any integers $a > b \geq 1$ we have

$$T(a, b) < \log_{\varphi}(\sqrt{5}b).$$

Proof We start with Binet's formula for Fibonacci numbers:

$$F_n = \frac{\varphi^n - \varphi^{-n}}{\sqrt{5}} > \frac{\varphi^n - 1}{\sqrt{5}}.$$

Let $k = T(a, b)$. By Lamé's Theorem, $b \geq F_{k+1}$. Substitute the lower bound of F_{k+1} :

$$b \geq F_{k+1} > \frac{\varphi^{k+1} - 1}{\sqrt{5}} \implies \sqrt{5}b + 1 > \varphi^{k+1}.$$

The function $\log_{\varphi}(\cdot)$ is strictly increasing, so take logarithm base φ on both sides:

$$\log_{\varphi}(\sqrt{5}b + 1) > k + 1 > k.$$

Thus $T(a, b) = k < \log_{\varphi}(\sqrt{5}b + 1)$, completing the proof. ■

Theorem 24 Suppose N is an L -bit composite integer, and x satisfies $x^2 \equiv 1 \pmod{N}$ with $1 \leq x \leq N$. Then $\gcd(x - 1, N)$ and $\gcd(x + 1, N)$ can be calculated with $O(L^3)$ arithmetic operations.

Proof The Euclidean algorithm for two L -bit integers takes $O(L^2)$ bit operations per iteration, and the number of iterations is bounded by $O(L)$. ■

Success Probability of Valid Order r

Theorem 25 Let $N = pq$ and p, q are distinct odd primes. Choose an integer x uniformly at random such that $1 \leq x \leq N - 1$ and $\gcd(x, N) = 1$. Let $r = \text{order of } x \text{ modulo } N$. Then

$$\mathbb{P}(r \text{ is even and } x^{r/2} \not\equiv -1 \pmod{N}) \geq 1/2.$$

Proof By the Chinese Remainder Theorem, there exists a group isomorphism

$$(\mathbb{Z}/N\mathbb{Z})^\times \cong (\mathbb{Z}/p\mathbb{Z})^\times \times (\mathbb{Z}/q\mathbb{Z})^\times \cong \mathbb{Z}_{p-1} \oplus \mathbb{Z}_{q-1}.$$

Let $x \mapsto (\bar{k}, \bar{l}) \in \mathbb{Z}_{p-1} \oplus \mathbb{Z}_{q-1}$, then

$$r = \text{ord}_N x = \text{lcm}(\text{ord}_{p-1}(\bar{k}), \text{ord}_{q-1}(\bar{l})) = \text{lcm}(r_1, r_2).$$

$$r_1 = \frac{p-1}{\gcd(k, p-1)}, \quad r_2 = \frac{q-1}{\gcd(l, q-1)}, \quad r = \text{lcm}(r_1, r_2).$$

And since -1 is the unique element of order 2 in $\mathbb{Z}_N^\times$, we have

$$x^{r/2} \equiv -1 \pmod{N} \iff \frac{r}{2}(\bar{k}, \bar{l}) = \left(\frac{p-1}{2}, \frac{q-1}{2} \right).$$

Let $E = \{r \text{ is even and } x^{r/2} \not\equiv -1 \pmod{N}\}$, and v_2 the 2-adic valuation. Let $s = v_2(p-1) \geq 1$, $t = v_2(q-1) \geq 1$. By valuation property:

$$v_2(r_1) = v_2\left(\frac{p-1}{\gcd(k, p-1)}\right) = s - v_2(k), \quad v_2(r_2) = v_2\left(\frac{q-1}{\gcd(l, q-1)}\right) = t - v_2(l).$$

Recall $v_2(r) = \max\{v_2(r_1), v_2(r_2)\}$, so the condition " r even" is equivalent to $\max\{s - v_2(k), t - v_2(l)\} \geq 1$. Therefore

$$\mathbb{P}(E) = \frac{\#\left\{(\bar{k}, \bar{l}) : \max[s - v_2(k), t - v_2(l)] \geq 1, \frac{r}{2}(\bar{k}, \bar{l}) \neq \left(\frac{p-1}{2}, \frac{q-1}{2}\right)\right\}}{(p-1)(q-1)}.$$

Let $\bar{E}$ denote the complementary event of E , i.e.

$$\bar{E} = \{r \text{ odd}\} \cup \left\{r \text{ even and } \frac{r}{2}(\bar{k}, \bar{l}) = \left(\frac{p-1}{2}, \frac{q-1}{2}\right)\right\}.$$

The equality $\frac{r}{2}(\bar{k}, \bar{l}) = \left(\frac{p-1}{2}, \frac{q-1}{2}\right)$ implies

$$r(\bar{k}, \bar{l}) = (0, 0) \in \mathbb{Z}_{p-1} \oplus \mathbb{Z}_{q-1} \implies \begin{cases} rk = (p-1)n_1, \\ rl = (q-1)n_2 \end{cases}$$

for some integers n_1, n_2 . Divide by 2:

$$\frac{r}{2}(\bar{k}, \bar{l}) = \left(\frac{p-1}{2}n_1, \frac{q-1}{2}n_2\right).$$

This vector equals $\left(\frac{p-1}{2}, \frac{q-1}{2}\right)$ if and only if n_1, n_2 are both odd. Take 2-adic valuation on $rk = (p-1)n_1$, $rl = (q-1)n_2$:

$$\begin{cases} v_2(r) + v_2(k) = v_2(p-1) = s, \\ v_2(r) + v_2(l) = v_2(q-1) = t. \end{cases}$$

This system is the necessary and sufficient condition for the second part of $\bar{E}$. We split into four cases by parity of k, l .

- Case 1: k odd, l odd. Then $v_2(k) = v_2(l) = 0$, the system reduces to

$$v_2(r) = s, \quad v_2(r) = t.$$

Since $v_2(r) = \max\{s, t\}$:

- If $s = t$: the system admits solution, so $\mathbb{P}(E \mid k, l \text{ odd}) = 0$;
- If $s \neq t$: no solution exists, so $\mathbb{P}(E \mid k, l \text{ odd}) = 1$.
- Case 2: k odd, l even. Here $v_2(k) = 0$, $v_2(l) \geq 1$, system becomes

$$v_2(r) = s, \quad v_2(r) + v_2(l) = t \implies v_2(l) = t - s.$$

- If $t \leq s$: $t - s \leq 0$, no such integer l , so $\mathbb{P}(E \mid k \text{ odd}, l \text{ even}) = 1$;
- If $t \geq s + 1$: the density of l with $v_2(l) = t - s$ is $1/2^{t-s} \leq 1/2$, so $\mathbb{P}(E \mid k \text{ odd}, l \text{ even}) \geq 1 - 1/2 = 1/2$.
- Case 3: k even, l odd. Symmetric to Case 2: $v_2(l) = 0$, $v_2(k) \geq 1$,

$$v_2(r) = t, \quad v_2(r) + v_2(k) = s \implies v_2(k) = s - t.$$

- If $t \geq s$: $s - t \leq 0$, no such k , $\mathbb{P}(E \mid k \text{ even}, l \text{ odd}) = 1$;
- If $t \leq s - 1$: $\mathbb{P}(E \mid k \text{ even}, l \text{ odd}) \geq 1 - 1/2^{s-t} \geq 1/2$.
- Case 4: k even, l even. $\mathbb{P}(E \mid k, l \text{ even}) \geq 0$.

Now all four parity pairs are equally likely, each with probability $1/4$. Consider the worst case $s = t$, which minimizes $\mathbb{P}(E)$:

$$\begin{aligned} \mathbb{P}(E) &= \frac{1}{4} \cdot \mathbb{P}(E \mid k, l \text{ odd}) + \frac{1}{4} \cdot \mathbb{P}(E \mid k \text{ odd}, l \text{ even}) + \frac{1}{4} \cdot \mathbb{P}(E \mid k \text{ even}, l \text{ odd}) + \frac{1}{4} \cdot \mathbb{P}(E \mid k, l \text{ even}) \\ &\geq \frac{1}{4} \cdot 0 + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 0 = \frac{1}{2}. \end{aligned}$$

For arbitrary s, t , at least two of the four cases have conditional probability 1, so the lower bound $\mathbb{P}(E) \geq 1/2$ always holds. ■

A Critical Flaws in Shors Algorithm Proofs of Nielsen & Chuang

This appendix records three substantial mathematical errors in the Shors algorithm chapter of *Quantum Computation and Quantum Information*. All defects are structural logical gaps rather than typographical slips.

A.1 Incorrect lower bound of factoring success probability

Let $N = \prod_{i=1}^m p_i^{\alpha_i}$ be an odd composite integer, where m counts distinct prime factors of N . Sample uniform $x \in [2, N-1]$ with $\gcd(x, N) = 1$, and let $r = \text{ord}_N(x)$. Define the successful event

$$\mathcal{E} = \{r \text{ even}, x^{r/2} \not\equiv -1 \pmod{N}\}.$$

The textbook incorrectly claims

$$\mathbb{P}(\mathcal{E}) \geq 1 - \frac{1}{2^m}.$$

The rigorous lower bound via group theory and inclusionexclusion principle is

$$\mathbb{P}(\mathcal{E}) \geq 1 - \frac{1}{2^{m-1}}.$$

A.1.1 Explicit counterexample verification with $N = 21 = 3 \times 7, m = 2$

$N = 21$ has two distinct prime factors, so $m = 2$.

1. Compute Euler's totient function: $\phi(21) = \phi(3)\phi(7) = 2 \times 6 = 12$. The multiplicative group modulo 21 is

$$\mathbb{Z}_{21}^\times = \{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}.$$

2. Textbook predicted lower bound:

$$1 - \frac{1}{2^m} = 1 - \frac{1}{4} = \frac{3}{4}.$$

If the books formula were true, at least $12 \times \frac{3}{4} = 9$ elements should satisfy event $\mathcal{E}$.

3. Enumerate failure condition: $x \in \mathbb{Z}_{21}^\times$ fails if its order r is odd, or r even with $x^{r/2} \equiv -1 \pmod{21}$.

- Compute order r for every element and test the failure rule:

- Fail elements: 1, 8, 13, 20, 5, 16 (total 6 elements).

- Successful elements satisfying $\mathcal{E}$: remaining 6 elements.

4. True probability calculation:

$$\mathbb{P}(\mathcal{E}) = \frac{\text{number of successful } x}{\phi(21)} = \frac{6}{12} = \frac{1}{2}.$$

5. Comparison:

- Corrected bound: $1 - \frac{1}{2^{m-1}} = 1 - \frac{1}{2} = \frac{1}{2}$, exactly matches the true probability;

- Textbook bound $\frac{3}{4}$ is strictly larger than the actual probability $\frac{1}{2}$, which proves $1 - 1/2^m$ is an invalid lower bound.

A.2 Misstated coprimality condition for order reconstruction via $\text{lcm}(r_1, r_2)$

Two independent quantum order-finding runs output reduced convergents $\frac{s_1}{r_1}, \frac{s_2}{r_2}$ after continued fraction simplification, where s_1, s_2 are simplified numerators. **The textbook claims $\text{gcd}(s_1, s_2) = 1$ is sufficient to guarantee $\text{lcm}(r_1, r_2)$ equals the true order r .**

Let s', s'' denote raw phase shift residues from phase estimation equations $2^n \varphi \equiv s \pmod{r}$ before fraction reduction. The valid sufficient condition is

$$\text{gcd}(s', s'') = 1,$$

not the reduced convergent numerators s_1, s_2 . The textbook confuses simplified fraction numerators with unprocessed modular phase offsets, breaking the algebraic logic for full order recovery.

A.2.1 Counterexample illustration

Suppose the raw phase fractions obtained from two independent phase estimations are both $\frac{50}{100}$.

- Raw residues: $s' = 50, s'' = 50$, so $\text{gcd}(s', s'') = 50 \neq 1$.

After simplification via $\text{gcd}(50, 100) = 50$, both convergents reduce to $\frac{1}{2}$:

- Reduced numerators: $s_1 = 1, s_2 = 1$, giving $\text{gcd}(s_1, s_2) = 1$.

Following the textbooks wrong rule, $\text{gcd}(s_1, s_2) = 1$ would incorrectly imply $\text{lcm}(r_1, r_2) = r$. In reality, the raw residues share a large common divisor, which means the candidate orders $r_1 = r_2 = 2$ only capture a divisor of the true order and cannot recover the full r . This example shows coprimality of simplified numerators carries no information about the original phase offsets; only $\text{gcd}(s', s'')$ controls whether $\text{lcm}(r_1, r_2)$ recovers the complete multiplicative order.

References

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